

Weekly Progress Report

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Domain: CORE JAVA

Date of Submission: 30-05-2026

Week Ending: 02

I. Overview

This week, the primary focus shifted towards active development of the Banking Information System project. Significant progress was made in implementing core modules and refining existing features. Alongside project development, efforts continued to strengthen Core Java fundamentals through dedicated learning and practice.

II. Achievements

1. Project Development — Banking Information System

- Actively developing key modules of the Banking Information System, including account management and transaction processing features.
- Implemented core business logic for customer data handling and account operations.
- Integrated input validation and exception handling to ensure application stability.
- Reviewed and refactored Week 1 code contributions for better readability and maintainability.

2. Java Concepts Learned

- **Object-Oriented Programming (OOP):** Deepened understanding of classes, objects, inheritance, encapsulation, and polymorphism as applied in the project.
- **Exception Handling:** Learned to use try-catch-finally blocks and custom exceptions to handle runtime errors gracefully.
- **Collections Framework:** Explored ArrayList, HashMap, and LinkedList for managing banking records and transaction data efficiently.
- **File I/O:** Studied Java file handling concepts to understand data persistence mechanisms.
- **String Manipulation:** Practiced various String methods and StringBuilder for processing user input and output formatting.

3. Collaboration and Communication

- Actively collaborated with team members to align on project structure and coding standards.
- Participated in code review sessions and incorporated feedback from mentors.

III. Challenges

4. Project Complexity

- Encountered difficulties in designing the class hierarchy for the Banking Information System to ensure scalability and clean separation of concerns.
- Managing data flow between multiple modules required careful planning and iteration.

5. Applying New Java Concepts

- Integrating newly learned Collections Framework concepts into the existing project codebase presented initial challenges.
- Debugging logic errors in exception handling blocks required additional time and attention.

IV. Learning Resources

6. Java Learning Platforms

- Referred to official Java documentation and online tutorials to understand OOP principles and Collections Framework.
- Followed video lectures and coding exercises on Java learning platforms to reinforce concepts.

7. Practical Application

- Applied learned concepts directly within the Banking Information System project for hands-on reinforcement.
- Practiced coding challenges related to exception handling and data structures for skill improvement.

V. Next Week's Goals

8. Project Development

- Continue developing and completing pending modules of the Banking Information System.
- Begin writing unit tests to validate implemented features.
- Seek mentor feedback to identify areas of improvement in the codebase.

9. Java Learning Goals

- Explore Java interface and abstract class concepts for better project design.
- Study JDBC basics for integrating database connectivity into the Banking Information System.

VI. Additional Comments

Week 2 has been a productive and learning-intensive period. The transition from familiarization to active project development has been smooth, thanks to the foundational knowledge built in Week 1. The team environment has been supportive, and mentor guidance continues to play a key role in navigating project complexities. Looking forward to making further progress in the coming week.

Submitted by: NAGESHWARAN C | Week 02 | 30-05-2026